



CSE4078S26 Final Report

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1 Abstract

This report explains what we did in our project of Turkish medical question answering LLM and we will compare base model with fine-tuning, starting from base model and reaching to evaluation the model after fine-tuning, focussing in modicle domain by usnig (alibayram/doktorsitesi) as SFT data. Splitting the data in two parts for training and testing. Quantitative results show that the answer length of characters reduces to -73% while imporving all ROUGE metrixes episall in ROUGE-2 which imporve be 148% after fine-tuning

2 Introduction

The goal of this project is to build LLM for Turkish language and then analyze it's performance. This report documents the fine-tuning of Qwen base model using LoRA on the alibayram/doktorsitesi SFT dataset. While building the LLM we insure that the test data is unseen in the training phase, by using 1,500 patient questions. Then we analysis the outputs of fine-tuning and base model by using ROUGE metrics and qualitative analysis.

3 Why Qwen 2.5 (1.5B) is better?

3.1 Qwen 2.5 (1.5B)

- Strong Turkish language support.
- Good reasoning for its size.
- Efficient on weak GPUs.

3.2 Phi-3.5-mini (3.8B)

- Strong reasoning.
- Larger context window.
- Competitive with much larger models.

While Phi-3.5-mini has more parameters, we find it very hard to train and to customize. Rather than this some times the Phi-3.5-mini produce very bad

answers as it appear in file (Qwen vs Phi.ipynb) in results folder, so me select (Qwen 2.5).

4 Experiment Setup

4.1 Base Model

- Model Architecture: Qwen 2.5 (small LLM)
 - Model number of parameters: 1.5B parameters.
 - Initial State: Pre-trained, no fine-tuning on medical data
 - Output Source: qwen_base_outputs_full.csv (1,500 test examples)
- Note:** all output files are available in our github, under results folder.

4.2 Fine-Tuning Method

- Method: LoRA (Low-Rank Adaptation)
- SFT Data: alibayram/doktorsitesi training split only
- Test Split: Completely unseen during training (no data leakage)
- Fine-tuned Model Output Source: qwen_lora_outputs_full.csv (1,500 test examples)

4.3 Evaluation Dataset

The evaluation corpus consists of 1,500 test examples from the test split. The data covers 9 Turkish medical specialties:

Each test example contains 5 columns:

- A specialty (kategori)
- A patient question (soru)
- A reference answer from a medical professional (reference_cevap)
- The base model's response (base_answer)
- The LoRA-fine-tuned model's response (fine_tuned_answer)

Specialty	Question Count
Beyin ve Sinir Cerrahisi (Neurosurgery)	384
Üroloji (Urology)	314
Ortopedi ve Travmatoloji (Orthopedics)	205
Genel Cerrahi (General Surgery)	182
Kulak Burun Boğaz (ENT)	143
Fiziksel Tıp ve Rehabilitasyon (PM&R)	133
Kardiyoloji (Cardiology)	72
Kalp Damar Cerrahisi (Cardiovascular Surgery)	38
Dahiliye / İç Hastalıkları (Internal Medicine)	29

4.4 Evaluation Metrics

- ROUGE Scores (ROUGE-1, ROUGE-2, ROUGE-L, ROUGE-Lsum): Measure n-gram overlap between model outputs and reference doctor answers. Higher scores indicate better alignment with expert responses.
- Average Answer Length: Assesses conciseness and whether fine-tuning reduces verbosity.
- Comparative Analysis: Direct comparison of base vs. fine-tuned outputs on identical test questions.

5 Experiment Results

5.1 Quantitative Results

The following results are from comparison_summary.csv in results folder, comparing base and LoRA-fine-tuned models on the 1,500-question test set:

Key Findings:

- LoRA fine-tuning reduce the answer length dramatically from 491 to 130 characters (about -73%), prducting short answer that match the reference ansers length (about 185 characters).

Metric	Base Model	LoRA-Fine-Tuned	Change
Avg. answer length (chars)	491.04	130.11	-73%
ROUGE-1	0.094	0.108	14%
ROUGE-2	0.012	0.03	148%
ROUGE-L	0.065	0.089	37%
ROUGE-Lsum	0.073	0.09	24%

- All ROUGE metrics improved dramatically. ROUGE-2 reach 148% better than the base model, which mean that the model start generate more structurally responses which match the reference doctor answers.
- The base model's generate very long answer, nearly 3x longer than reference answer, his answers were reflecting verbose and rambling outputs.

5.2 Qualitative Observations

Base Model Characteristics:

- Verbose and rambling: Answers average 491 characters — nearly 3x longer than reference answers.
- Often incoherent: many times the model produce repetitive phrases and occasionally mixes in non-Turkish characters.
- The model show poor alignment with references: rather than the length, the model often fails to capture the correct key clinical message.

LoRA-Fine-Tuned Model Characteristics:

- Concise and correct: The length of characters is 130 in average which is close to reference answer length.
- Clinically appropriate: Clear and actionable answers and recommendations.
- Effective specialist: The fine-tuning model generate Frequently and correct directs patients to the appropriate specialists.

- Strong alignment: ROUGE-2 score confirms structural similarity to doctor answers.

5.3 Example Comparison

Aspect	Base Model	LoRA-Fine-Tuned	Reference
Length	~400-600 characters	~100-150 characters	~185 characters
Style	Rambling, bullet points, repetitive	Concise, direct, professional	Professional
Actionability	Often vague	Clear next steps	Clear next steps
Clinical alignment	Poor	High	—

6 Analysis

6.1 Does Fine-Tuning Improve Performance?

Yes it is. The fine-tuning model outperforms the base model on all evaluation metrics:

- 73% reduction in length, meaning successful learning conciseness.
- 148% improvement in ROGUE-2, which demonstrates much better structural alignment with the doctor answers.
- all ROGUE metrics get higher scores than the base model. While we split the data to training and test datasets, The ROGUE scores means that our fine-tuning model start generate genuine generalization, not memorization.

6.2 Why LoRA Was Effective?

LoRA's parameter-efficient approach was well-suited to this task for several reasons:

- Computational efficiency: Full fine-tuning for the whole of model would be prohibitive; LoRA made the project manageable.
- Domain adaptation without catastrophic forgetting: Using LoRA the base model preserved his capabilities while adapting to medical domain patterns.
- Turkish language adaptation: The SFT data provided rich Turkish medical

language that LoRA effectively incorporated.

6.3 Error Patterns

Based on review of the fine-tuned model outputs:

- Occasional over-conciseness: Some times the model too brief for complex cases.
- Specialty referral accuracy: Generally correct. However, there are some misdirections.
- Overall error rate: Substantially lower than the base model

LoRA fine-tuning model solve many errors in General, such as rambling, incoherence and off-target responses.

7 Conclusion

This project successfully demonstrates that LoRA fine-tuning on domain-specific SFT data (alibayram/doktorsite) significantly improves small LLM performance on Turkish medical QA tasks. Key Findings:

1. LoRA fine-tuning is highly effective for Turkish medical domain, producing 148% improvement in ROGUE-2 and a 73% reducing the length answer.
2. The fine-tuning model generalizes the unseen test data, which confirm the improvement due to learning, not memorizing.
3. Parameter-efficient fine-tuning made the project manageable while delivering substantial performance gains.

The fine-tuned model is clear effective and practical for Turkish medical QA deployment that the base model.

In the end, this project help us to understand how to choose and select base model for specify domain, and then how to fine-tune the model and finally how to evaluate the model to check it performance.